



Deciphering groundwater quality, mechanisms controlling groundwater chemistry in and around Suryapet, Telangana, South India

K. Saikrishna^a, D. Purushotham^b, V. Sunitha^{c,*}, Y. Sudharshan Reddy^d, T. Brahmaiah^e, B. Muralidhara Reddy^f, B. Nallusamy^g

^a Department of Geology, Kakatiya University, Warangal, Telangana 506009, India

^b CSIR-National Geophysical Research Institute, Hyderabad, Telangana 500007, India

^c Department of Geology, Yogi Vemana University, Kadapa 516005, India

^d Department of Civil Engineering, Sai Rajeswari Institute of Technology, Proddatur, Kadapa, AP 516362, India

^e Mineral Exploration Corporation Limited, Nagpur, Maharashtra 440006, India

^f Department of Earth Science, Yogi Vemana University, Kadapa, Andhra Pradesh 516005, India

^g Department of Geology, Central University of Karnataka, Karnataka 585367, India

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ABSTRACT

The main objective of the present paper is to determine the groundwater quality in and around Suryapet district, Telangana, and also to define groundwater chemistry mechanisms. For this, groundwater samples from thirty locations were analyzed for physicochemical parameters such as pH, EC, TDS, TH, major cations, and anions. Most of the samples were beyond the World Health Organization (WHO, 2011) maximum permitted levels for drinking. Groundwater is alkaline in nature, very hard. The abundance of the cations and anions is $\text{Ca}^{2+} > \text{Na}^+ > \text{Mg}^{2+} > \text{K}^+$; $\text{Cl}^- > \text{HCO}_3^- > \text{CO}_3^{2-} > \text{SO}_4^{2-} > \text{NO}_3^- > \text{F}^-$ respectively. The major ion distribution is regulated by cation exchange and silicate weathering processes; Gibbs plots show that the evolution of groundwater geochemistry depends on the interaction of water–rock followed by the evapotranspiration mechanism. Cation-anion exchange and base-exchange reactions play a dominant role. Depending on the factor analysis results, four factors are contributing to 74 percent of groundwater contamination and spatial variation maps decipher nonionic and ionic concentration due to geogenic and anthropogenic activities. This study suggests that contaminated groundwater should be treated before being consumed by humans.

Introduction

In India owing to paucity of surface waters, majority of people depend on groundwater for several purposes like drinking and domestic, industrial, and irrigation. Therefore supply of safe drinking water is very essential for human development (World Health Organization (WHO), 2011). Groundwater is the supreme natural resource and precious national asset; it has attained an important position in overall water resource. However due to rapid urbanization and industrialization, expanding population, emerging trends in technology, depletion of groundwater resources causes environmental pollution and health effects. Due to lack of proper waste management strategies, several anthropogenic activities and their by-products have produced pollution of surface and subsurface water. Effluents from industries, agriculture, horticulture transport all ultimately exert influence on water quality. Hence quality of drinking water is very crucial in various

health issues. These problems vary widely and rely on geology, hydrogeology, climate and geochemical mechanisms. Quality and quantity of recharged water, geologic structure, aquifer mineralogy are the significant components that contribute chemical quality of groundwater (Yousefi et al., 2018). The hydrochemical study reveals Groundwater quality relies on lithological, pedogeochemical compositions, human activities and several geochemical compositions of the rocks (Adimalla and Taloor, 2020). Lithology, geochemistry can alter ground water hydro chemical features further through subsurface movement groundwater chemistry can also be altered through different anthropogenic sources like waste disposal practices, hygienic conditions (Brindha and Elango, 2015; Sunitha and Reddy, 2019; Sunitha et al., 2022; Sudharshan Reddy et al., 2020a). Understanding groundwater quality and its mechanism is very essential for effective management and sustainable resource for various purposes. Groundwater chemistry relies on many geochemical reactions along the flow path, so it is very

* Corresponding author.

E-mail address: vangalasunitha@gmail.com (V. Sunitha).

important to locate the aquifer concerning to determine the dispersal and mechanism of the region's main ion chemistry (Fisher and Mullican, 1997; Rosen and Jones, 1998; Singh, 2000). A region's groundwater chemistry is typically not homogeneous and is influenced by flow, geochemical processes, evaporation and evapotranspiration and potential pollution sources (Sunitha et al., 2012; Sunitha et al., 2022; Sreedevi et al., 2018). Recognition of several related geochemical processes will help to understand the causes of changes in water quality due to aquifer contact, especially in weathered rock formations. Hydrogeochemical processes can also help to prepare and maintain polluted sites in order to preserve aquifers contaminated by natural and anthropogenic phenomena. Knowledge of geochemical processes that regulate groundwater chemistry is therefore important for understanding and dealing with groundwater quality issues (Sreedevi et al., 2019a,b). Geochemical techniques for ion exchange, oxidation–reduction, dissolution, precipitation, minerals weathering, and groundwater habitation of time all have an impact on the composition of groundwater (Rina et al., 2013). In addition, human activities such as increased pumping and release of contemporary effluents have successfully modified the formation of groundwater (Hamilton and Helsel, 1995).

Groundwater pollution is caused by many factors, such as geogenic or anthropogenic, nitrogen pollution, agricultural activities, rock and soil dissolution, mine water contamination, according to previous literature (Brindha and Elango, 2015; Adimalla and Venkatayogi, 2017; Sreedevi et al., 2017; Saikrishna et al., 2020, 2021a,b). Several researchers have found that groundwater quality studies have been analysed, viz., (Satyanarayanan et al., 2007; Arumugam and Elangovan, 2009; Belkhihi and Mouni, 2012; Purushotham et al., 2011; Pradeep Kumar and Srinivas, 2012; Moosavirad et al., 2013; Arunprakash, et al., 2014; Singh et al., 2015; Shabya Choudhary et al., 2016; Mohammed et al., 2017; Syed and Umair, 2018).

This region's groundwater is applied for irrigation, domestic and drinking water needs. It is necessary for the sustainable development of water supplies for drought locations to understand the geochemical evolution of groundwater, as there are restricted sources for the aquifer's susceptibility to contamination. In many regions of the world, the importance of groundwater hydrochemistry has prompted a number of thorough research on the deterioration of groundwater quality and the geochemical development of groundwater (Purushotham et al., 2011; Sunitha et al., 2022; Kadam et al., 2021). Numerous scholars from all around the world have examined groundwater hydro-chemical data in order to use chemometric approaches to analyze the source and history of ions. Numerous research on the geochemistry of semiarid region waters, including the suitability of the water for drinking and irrigation, have been published (Mgbenu and Egbueri, 2019; Zhang et al., 2020). Numerous studies on groundwater quality assessment for drinking and irrigation purposes have been carried out in India, especially in Andhra Pradesh, such as; Sudharshan Reddy et al., 2020b; Adimalla and Qian, 2019; Sunitha and Reddy, 2019; Narsimha and Sudarshan, 2018; Subba Rao et al., 2016; Purushotham et al., 2017; Narsimha and Sudarshan, 2017a,b, Sreedevi et al., 2016.

In general, groundwater chemistry of a region is not uniform, and it is influenced by flow, geochemical processes, evaporation and evapotranspiration, as well as potential pollution sources. Understanding the reasons of water quality changes brought on by contact with an aquifer, particularly in worn rock formations, would be made easier with the recognition of various linked geochemical processes. Understanding the reasons of water quality changes brought on by contact with an aquifer, particularly in worn rock formations, would be made easier with the recognition of various linked geochemical processes. Aquifers that have been contaminated by both natural and man-made occurrences can be preserved by using hydrogeochemical techniques to clean up and sustain polluted locations. Understanding the geochemical processes that control groundwater chemistry is crucial for resolving problems with groundwater quality (Sreedevi et al., 2019a,b;

Sunitha and Reddy, 2019; Subba Rao et al., 2018). The major findings of this study are based on the integration of hydrogeological and hydrochemical data from the groundwater system.

Groundwater chemistry in the region is regulated by a number of causes and mechanisms. To comprehend quality degradation and to plan corrective actions to prevent future harm, periodic water quality monitoring is necessary. Groundwater of this region is used for irrigation, domestic and drinking water needs. It is necessary for the sustainable development of water supplies for semi-arid regions to understand the geochemical evolution of groundwater, as there are restricted sources for the aquifer's susceptibility to contamination. Hence the study focused on the main objectives of (1) to evaluate the chemical components of groundwater (2) to identify the mechanisms governing its groundwater chemistry of the region. This research work could aid in the hydrochemical interpretation of new researchers and governing bodies for the sustainable management and protection of groundwater resources in the Suryapet and neighbouring areas of Telanagana, state of Andhra Pradesh.

Study area

The field of study falls between 79°35'30" and 79°40'35" E longitude and 17°05'00" and 17°10'00" N latitude (Fig. 1). This area is surrounded by hard rock terrain where excessive groundwater withdrawal occurs because of many anthropogenic activities (Brindha et al., 2010). Most of the study area consists of crystalline Archaean rocks such as granite, granite gneiss and schists. Limestones and quartzites belonging to the Cuddapah and Kurnool systems are found in the southern part of the study region. Along the rivers and streams, unconsolidated soils of alluvial sands and clay exist. The study area stretches over 81 km² and occupied with Granites, gneisses, schists, and intrusive make up the 90% of the district's Archaean crystalline rocks. In the southern part of the district, 9% of the area is made up of the consolidated metasedimentary rocks of the Cuddapah and Kurnool system, which include limestones, quartzites, and shales. Unconsolidated deposits, which constitute alluvial sands and clay, are distributed along the major rivers and streams and comprise around 1% of the total area. The granite greenstone terrain of EDC represents the Precambrian domain, the Proterozoic Cuddapah basin, and the Eastern Ghat Granulite Belt (Ramakrishnan and Vaidyanadhan, 2008). The EDC contains TTG (3.0 Ga) remnants and is characterised by greenstone belts with tholeiitic to calc-alkaline bimodal volcanic assemblages and lesser komatiite-tholeiite sequences representing intraoceanic arc magmatism and plume-derived oceanic plateau accretion and low potassium oxide concentration oceanic type of original magma were observed (Manikyamba et al., 2012; Card, 1990). The rocks have undergone structural disturbances due to tectonic activity and are subjected to metamorphism along with profuse granitic magma invasion. Subsequent intrusion of granites and gneisses by dolerite dykes pegmatite veins and quartz is also noticed. Economic minerals such as iron, copper manganese, lead, zinc, gold, magnetite, ilmenite, rutile, zircon and apatite Al-rich clay minerals like vermiculite and chlorite, Sulfide minerals etc were observed in this area. The area falls under semi-arid and tropical climate, the annual average temperature was 36 °C to 40 °C and the annual precipitation varied from 652 to 891 mm (average 798.1 mm) in 2019. Depth to water levels (DTW) ranges from 0.8 to 40 m below ground level (m bgl) (average: 7.10 m) and 0.15–55.9 m bgl (average: 7.6 m) during pre and post monsoon seasons and ground water yield varies from < 0.1 to 7.5 lps (avg: 1.83 lps). The NW-SE trending study region as a part of the Musi River sub-basin and Halia river sub-basin (Lower Krishna) of the Krishna river basin spanning in the Suryapet and Nalgonda district, Telangana State of southern India. Geomorphologically, the area is comprised of Pediplain which is the major land form, the other landforms observed are pediment, denudation hill, flood plain, residual

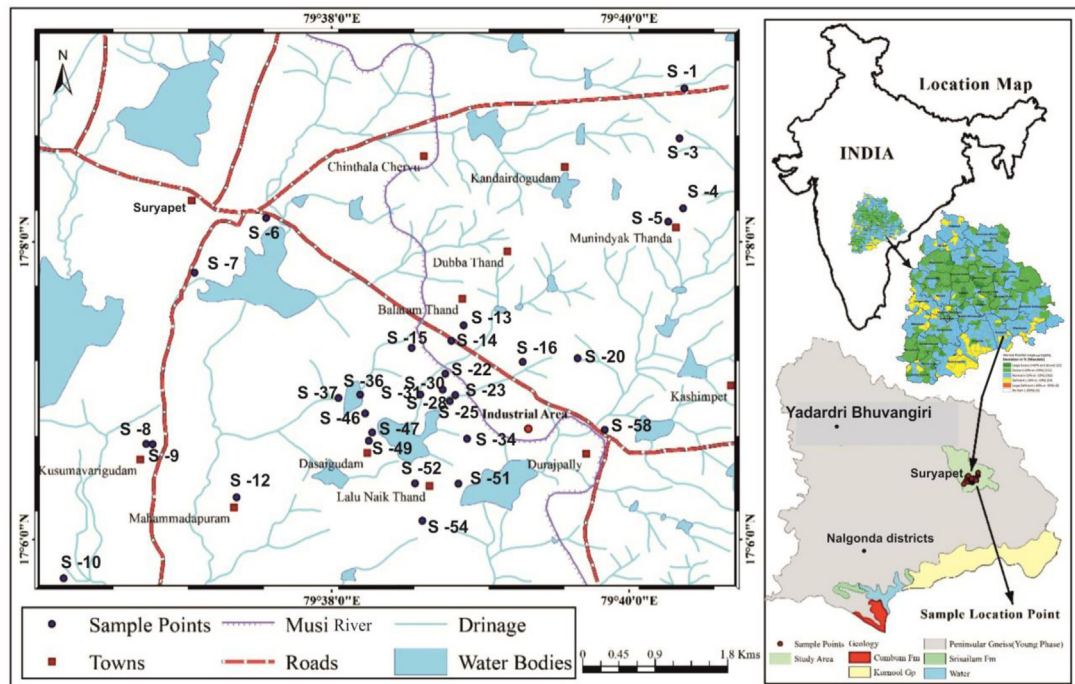


Fig. 1. Location map of the study area.

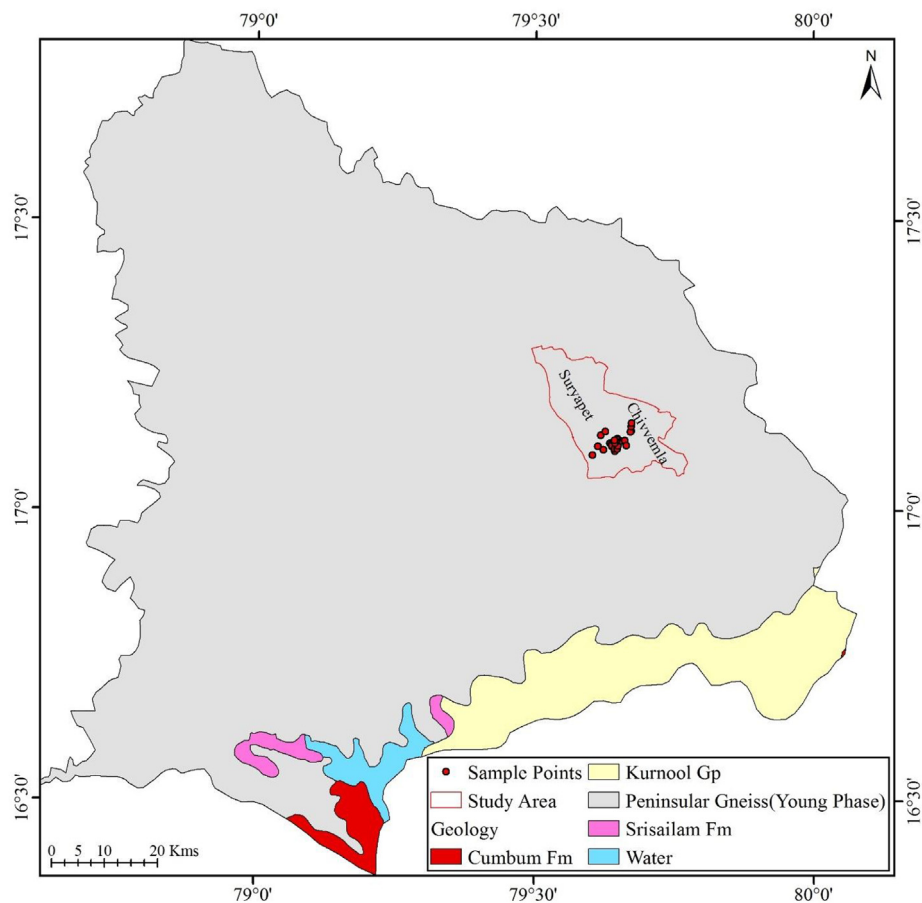


Fig. 2. Geological map of the study area.

hill, channel fill, etc. The drainage is controlled by lineaments trending NW-SE, EW, NE-SW and N-S directions and are drained by many streams with rivulets having sub-dendritic drainage (Fig. 2). In Musi sub-basin the general trend of lineament follows the NW-SE direction, whereas in Halia basin the trend of lineaments is NS and NW-SE direction. The area is mostly occupied by loamy, gravelly loam and gravelly clay soils. The cropping pattern of the study area during khariff is paddy, cotton, jowar, arhar, maize, bajra, sesame and chillies and during rabi season is paddy, horse gram, maize. The crystalline rocks lack of primary porosity, but latterly, the dynamic weathering process causes the rocks to fracture, fissure, and form joints. This results in the creation of secondary porosity, which serves as a reservoir for ground water. In the weathered zone, the ground water is found under water table conditions, while in the fractured zone, it is found under semi-confined and confined conditions. The depth ranges of Open-dug wells with 6 to 15 m and dug-cum-bore wells with a depth of up to 60 m are consider as the phreatic aquifer. Irrigation well yields range from 100 to 150 cu.m. per day. Sometimes, it can reach 200 cu. m/day (CGWB, 2013).

Materials and methods

Sampling and chemical analysis

Groundwater samples from thirty locations were obtained from various wells from shallow to deep aquifers (~10 to 40 m deep) during the post-monsoon season in 2019. For collecting samples, a Global Positioning System (GPS) instrument is used. Several physicochemical parameters determined were pH, electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH), calcium (Ca^{2+}), magnesium (Mg^{2+}), sodium (Na^+), potassium (K^+), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), chloride (Cl^-), sulfate (SO_4^{2-}), nitrate (NO_3^-) and fluoride (F^-). Immediately after sampling, pH, electrical conductivity (EC), and total dissolved solids (TDS) were measured in the field itself. Ion Chromatography is employed to analyze the cations and anions. By using a spectrophotometric technique, SO_4^{2-} and NO_3^- are identified. Using an ion selective electrode, F is calculated (Orion 4 star ion metre, Model: pH/ISE). A stock solution of sodium fluoride (100 mg/L) was used to create standard fluoride solutions (0.1–10 mg/L). To 20 ml of sample, 2 ml of grade III total ionic strength adjusting buffer (TISAB III) was added in accordance with the experiment's specifications. The standard procedures outlined in the standard methodology for the study of groundwater were used to analyze each parameter (Hem, 1991; Raghunath, 1987; APHA, 2012). Preservation of samples and analysis of water samples were followed as per the guidelines adopted by APHA (2012) and suggested precautions were taken to prevent contamination (Hem, 1991; Raghunath, 1987). All the tests were performed in triplicate and the precision is found to be within a $\pm 3\%$ error.

Analytical accuracy

In order to determine the analytical accuracy of total cation concentration (Na^+ , K^+ , Ca^{2+} , Mg^{2+}) and total anion concentration (CO_3^{2-} , HCO_3^- , Cl^- , SO_4^{2-} , NO_3^- and F^-) showed in milliequivalent per liter (meq/L) for individual sample, ionic balance error (IBE) was determined to ensure analytical accuracy by the equation expressed below:

$$\text{IBE} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100(\text{meq/L})$$

The computed IBE value is within admissible limit of $\pm 3.4\%$ (Domenico and Schwartz, 1990).

Results and discussions

Samples were tested for major cations and anions to determine the groundwater quality. The findings are reviewed in conjunction with

the BIS (2014) Bureau of Indian Standards and World Health Organization (WHO) (2011) World Health Organization drinking water standards (Table 1). Groundwater pH varies from 7.0 to 8.6 mg/L with an average of 7.81 suggesting alkaline activity (Only 3 percent of the samples are fall under desirable limit). The majority of the samples are at the desirable limit. In the study area, electrical conductivity (EC) reflects increased mineral content, salinity, and temperature. At 25 °C, the EC varies from 380 to 5910 $\mu\text{S}/\text{cm}$. As per the World Health Organization (WHO) (2011); BIS, (2014) criteria, ninety percent of the total samples are above the recommended level of $< 1000 \mu\text{S}/\text{cm}$. This EC variation may also be due to geochemical and anthropogenic processes in this area (Subba Rao, 2017; Muralidhara Reddy et al., 2013, 2015, 2016). Total dissolved solids (TDS) of groundwater with a mean of 1411 mg/L varies from 190 to 2960 mg/L (Table 1). Ninety percent of the groundwater is crossing the prescribed limit of 500 mg/L denoting non-potable category World Health Organization (WHO) (2011); BIS (2014). In spite of the maximum anthropogenic activities prevailing in this area, greater TDS content may be due to dissipation/precipitation mechanisms which facilitated the movement of groundwater via lineaments, fractures and joints (Krishna Kumar et al., 2014). The total groundwater hardness ranged from 180 to 2656 mg/L, ninety-six percent of groundwater as per BIS (2014) and World Health Organization (WHO) (2011) requirements is above the acceptable level of 300 mg/L.

Major ion chemistry

In small dosages, major ions are beneficial to humans, but in large doses, they can be toxic or even carcinogenic. The environment contains a variety of ions, which are characterized not only by their physicochemical forms but also by their differential toxicity to living things. With potentially detrimental environmental effects, water poisoning brought on by newly emerging pollutants is a growing global concern. Many different water resources contain newly emerging pollutants. Major ions make up a considerable portion of the total dissolved solids in ground water. The quantities of these ions in groundwater are determined by hydrogeochemical processes in the aquifer system (Basheer and Ali, 2018). However, geochemical reactions of the aquifer in order to have access to the distribution of the region's key ion chemistry. When the main ion concentration in groundwater approaches equilibrium, these events occur. As a result, the examination of concentrations of different main ions present in ground water is utilized to identify geochemical mechanisms. The key abundance of cations and anions in this location are as follows: $\text{Na}^+ > \text{Mg}^{2+} > \text{K}^+$ and $\text{Cl}^- > \text{HCO}_3^- > \text{CO}_3^{2-} > \text{SO}_4^{2-} > \text{NO}_3^- > \text{F}^-$. Calcium concentration ranges from 43.9 to 774.5 mg/L, and according to World Health Organization (WHO) (2011) and BIS (2014) standards, seventy percent of the total samples are above the prescribed limit of 75 mg/L. Majority of the groundwater samples are restricted to hard rock terrain, and due to rock-water interaction and mineral dissolution, higher Ca^{2+} in groundwater is observed. Sodium concentration ranged from 14.5 to 958.0 mg/L, and according to the World Health Organization (WHO) (2011) and BIS (2014) limits, eighty-five percent of the total samples are above the recommended range of 200 mg/L. In this area, higher Na^+ concentrations may be due to the leaching and decomposition of various igneous minerals. Other contributing sources may be due to the discharge of industrial and municipal waste discharge and agricultural sites (Muralidhara Reddy et al., 2019; Muralidhara Reddy and Sunitha, 2020). Magnesium concentration varies from 4.3 to 247.7 mg/L; as per World Health Organization (WHO) (2011) and BIS (2014) limits, eighty-seven percent of the total samples are above the usual level of 30 mg/L. According to World Health Organization (WHO) (2011) and BIS (2014) guidelines, potassium concentrations range from 0.9 to 87.5 mg/L and sixteen percent of the total samples surpass the recommended limit of 10 mg/L (Table 1). Cl^- ranges between 11.6 and 1925.50 mg/L. 75 percent

Table 1

Comparison of groundwater composition of present study area with drinking water quality Standards WHO (2004), BIS (2003) and ICMR (1975).

Parameter	Analyzed sample range	Mean	Standard deviation	WHO (2006)	BIS (2003)	Undesirable effects	% of samples exceeding the permissible limit
pH	7.22–8.6	7.81	0.30	7.0–8.5	6.5–8.5	Taste	3.33
EC $\mu\text{S}/\text{cm}$	630–5370	2631	560.42	1500	1500	–	87
TDS (mg/l)	197–2910	1411	1376.98	500	500	Gastro intestinal irritation	90
TH (mg/l)	180–2656	–	745.21	300	300	Poor lathering with soap and skin irritation	90
Ca^{2+} (mg/l)	43.9–774.5	159.61	144.02	75	75	Scale formation	75
Mg^{2+} (mg/l)	4.3–247.7	92.29	61.75	30	30	–	90
Na^+ (mg/l)	14.5–958.0	388.7	226.20	200	200	Hypertension and heart disease	85
K^+	0.9–87.5	8.42	16.71	10	10	Bitter taste	15
Cl^- (mg/l)	11.6–1925	656.7	523.77	250	250	Bitter taste	75
SO_4^{2-} (mg/l)	16.9726.4	188.45	157.29	200	200	Laxative effect	40
HCO_3^-	140–600	383	123.95	500	500	White deposit on fruits and leaves	15
CO_3^{2-}	0–80	27	30.84	–	–	–	–
NO_3^- (mg/l)	1.1–341.7	50.03	71.88	45	45	Blue baby	40
F^- (mg/l)	0.2–2.3	1.29	0.51	1.5	1.5	Fluorosis	50

of groundwater samples deviate from the normal prescribed limit prescribed by World Health Organization (WHO) (2011) and BIS (2014) of 250 mg/L. High groundwater chlorides can be attributable to weathering, rock and soil leaching, domestic and industrial effluent and dry atmosphere (Subba Rao, 2017). The concentration of sulphate ranged from 16.9 to 726 mg/L; 40% of groundwater exceeded the allowable level of SO_4^{2-} (World Health Organization (WHO), 2011; BIS, 2014). This may be attributed to the release of phosphate mineral (e.g. apatite) from weathering granites and even from domestic waste, industrial effluents, etc (Karanth, 1987; Craig and Anderson, 1979).

In addition, these variables may have led to salinity, which has exacerbated the soil in this area. Bicarbonate ranged from 140 to 600 mg/L, with 17% of groundwater above the appropriate 500 mg/L limit (World Health Organization (WHO), 2011; BIS, 2014). Carbonates varied from 0 to 80 mg/L which might be derived from carbon dioxide emitted by organic decomposition in the soil (Todd, 1980). The nitrate concentration varies from 1.1 to 341.7 mg/L, with 33% of the total groundwater samples above the prescribed 45 mg/L limit (World Health Organization (WHO), 2011; BIS, 2014). N-compounds from human, animal waste and agricultural appliances are the main sources of nitrates. The most commonly adopted fertilizers that account for nitrates in the groundwater of this area are urea $\text{CO}(\text{NH}_2)_2$ and ammonium nitrate $\text{NH}_4 \text{NO}_3$. Fluoride concentrations ranged from 0.2 to 2.3 mg/L, with 60% of the overall groundwater samples were above the recommended 1.5 mg/L limit of fluoride (World Health Organization (WHO), 2011; BIS, 2014). Though fluoride is good for human health in the prevention of dental caries and helps to sustain enamel growth for children under 8 years of age at 0.8–1.0 mg/L (Sunitha and Reddy, 2019); while groundwater with higher fluorides > 1.5 mg/L induces dental fluorosis and skeletal fluorosis when ingested in concentrations > 3.0 mg/L (Fig. 3(a–k)). There are three types of fluoride, namely fluorspar or calcium fluoride (CaF_2), apatite or rock phosphate ($\text{Ca}_3\text{F}(\text{PO}_4)_3$) and cryolite (Na_3AlF_6) (Sunitha et al., 2012). While groundwater contamination depends on different components such as temperature, pH, calcium and bicarbonate ion concentration in water, the main source of fluoride in groundwater is the weathering of granitic host rocks that causes the most prevalent leaching of fluoride bearing minerals in this area. Apart from these factors, seasonal, spatial and depth variations can also be the sources for fluctuations of varied fluoride concentration. Wide variety of water quality parameters in existence is essentially because of anthropogenic sources (Drever, 1987). Groundwater composition of this area is probably originated from silicate weathering process, sand disintegration/precipitation forms augmented by precipitation and groundwater flow. The sewage discharge of responsive contamination, farming practices, lacking of suitable land, water resources management practices, unplanned utilization of manures/pesticides, temporal

fluctuations and so forth are the fundamental causes of anthropogenic contamination.

Hydrogeochemical evaluation of Groundwater

In order to know and demonstrate the hydrogeochemical procedures intricated in the aquifer domain during the evolution of different groundwater facies, the concentration of several major elements and their correlation were observed. The solution of important cations such as Ca^{2+} , Mg^{2+} and alkaline metals and of important anions such as SO_4^{2-} , Cl^- , CO_3^{2-} and HCO_3^- may generally be described as natural water. Linear plots are also more accurate for denoting the composition of groundwater. Piper's Trilinear diagram 1944 consists of three different fields, two triangular fields and one diamond-shaped field. By their location in the diamond sector, groundwater can be recognized. The diagram of The Piper (1944) was applied to evaluate groundwater hydrogeochemical facies, groundwater characteristics and values, geochemical processes that exist within the aquifer. This plot offers a clear way to identify and relate various forms of water based on the chemical composition of groundwater (Hem, 1991). Most naturally occurring water can contain solutions of the three cationic elements magnesium, calcium, and sodium as well as the three anionic elements sulphate, chloride, and bicarbonate. Numerous researchers have suggested numerous trilinear charting types. Piper plot is applied to denote the categorization of water type, recognition of hydrogeochemical facies and mixing of two samples (Karmegam et al., 2011; Thivya et al., 2013). Piper's trilinear diagrams are both helpful for graphically illustrating the variations in main ion chemistry in groundwater flow systems. Based on the plots of major cations Ca^{2+} , Mg^{2+} , Na^+ and K^+ and major anions Cl^- , HCO_3^- and SO_4^{2-} content in the groundwater. Plotting the geochemical information on a Piper trilinear diagram allows for the analysis of the hydrogeochemical facies of groundwater and the interaction between the various dissolved ions. The Piper diagram has been extensively applied for identification of groundwater types which play a dominant role in groundwater chemistry. A point displayed in the cation triangle based on the percentage of three that are present indicates the composition of the system with respect to cations. The location of a comparable point in the anion triangle indicates the composition with regard to anions. The character of the water is revealed at the point where the extension contacts each point after projecting each one into the upper field along a line parallel to the upper field edge.

Piper diagram includes three prominent fields consisting of 1) Cations (Ca^{2+} , Mg^{2+} , Na^+ , K^+) triangular. 2) Anions (Cl^- , SO_4^{2-} , HCO_3^- , Cl^-) triangular. 3) Diamond shaped field (Fig. 7) Piper (1944). As per this classification, groundwater of this region can be grouped under $\text{Na}^+ - \text{HCO}_3^- - \text{Cl}^-$ and $\text{Na}^+ - \text{Cl}^- - \text{HCO}_3^-$ water

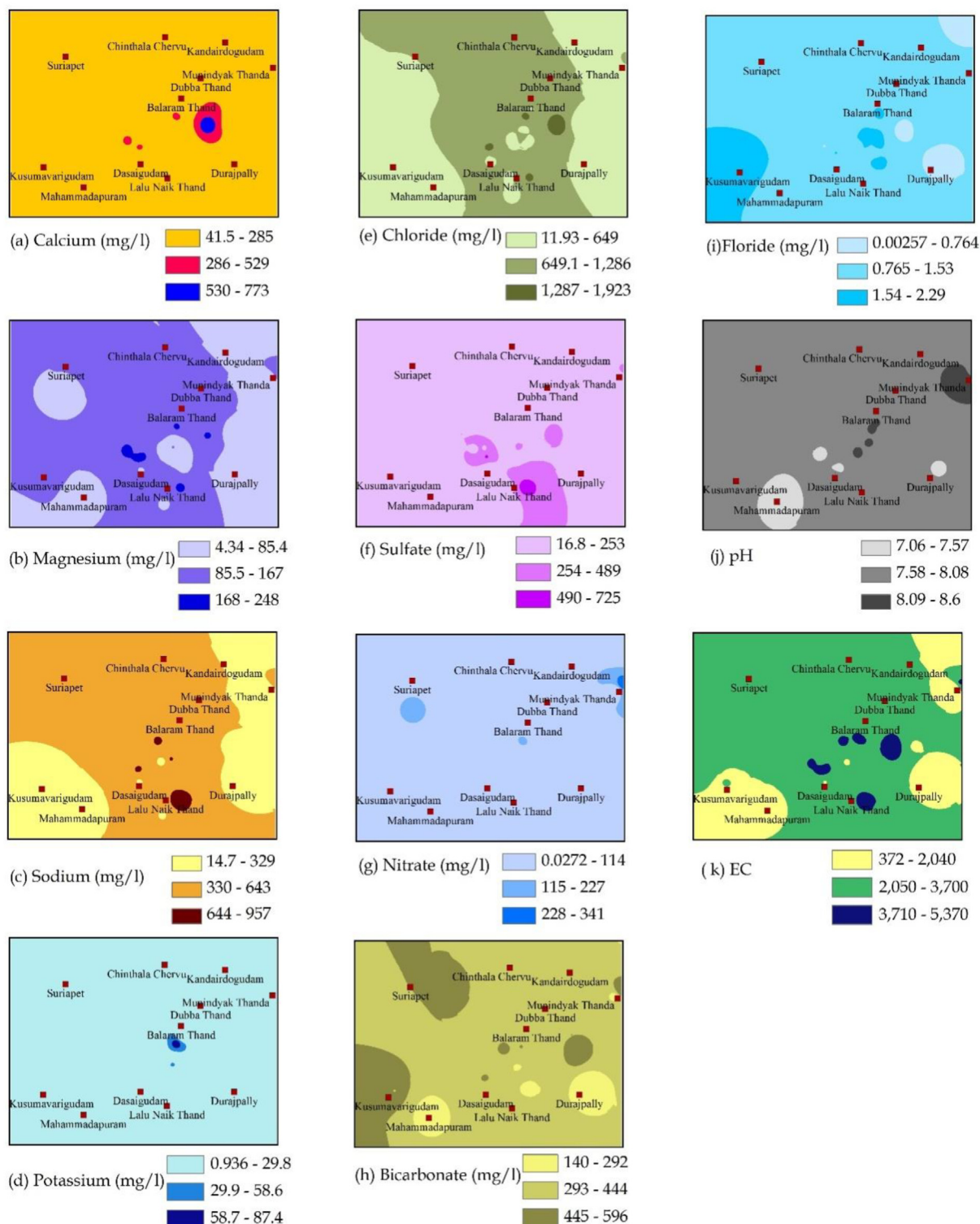


Fig. 3. a-k. Spatial variation maps of non ionic-ionic maps of the study area.

types (Table 3, Fig. 4). Sodium bicarbonate type of groundwater generally favours dissolution of fluoride (Handa, 1975). These outcomes were seen and are in coincidence with comparative perception as noted in Siddipet, Basara and Medak locales in Telangana, India, where higher fluoride concentration in groundwater is associated with high sodium bicarbonate (Sunitha et al., 2022; Narsimha and Sudarshan, 2017a,b).

Mechanism controlling the groundwater chemistry

The Gibbs (1970) diagram has been applied to understand the groundwater chemistry mechanism. It deals with chemical reaction using graphical plots between the water composition and the regulating mechanism of the variations in natural water chemistry. In addition to atmospheric precipitation, evaporation and water-rock

Table 2

Characterization of groundwater of the study area on the basis of Piper trilinear diagram.

Subdivision number of the diamond shaped field	Characteristics of corresponding sub division of diamond shaped fields	%of samples
1	Alkaline Earths ($\text{Ca}^{2+} + \text{Mg}^{2+}$) exceeds alkalies ($\text{Na} + \text{K}$)	Nil
2	Alkalies exceeds alkaline earths	Nil
3	Weak acids ($\text{CO}_3 + \text{HCO}_3^-$) exceeds Strong acids ($\text{SO}_4^{2-} + \text{Cl}^- + \text{F}^-$)	Nil
4	Strong acids exceed weak acids	Nil
5	Carbonate hardness (secondary alkalinity) exceeds 50% i.e. Chemical properties of the groundwater are dominated by alkaline earths and weak acids	13
6	Non carbonate hardness (secondary salinity) exceeds 50% i.e. Chemical properties of the groundwater are dominated by alkalis and strong acids	Nil
7	Non carbonate alkali (primary salinity) exceeds 50% i.e. Chemical properties of the groundwater are dominated by alkalis and weak acids	57
8	Carbonate alkali (primary alkalinity) exceeds 50% i.e. Chemical properties are dominated by alkalis and weak acids	NIL
9	No one cat ion – anion exceeds 50%	30

Table 3

Classification of water on the basis of Hydro-Chemical Facies.

Water Type	Percentage of groundwater
Cations	
Ca^{2+} type (A)	10
Mg^{2+} type (C)	Nil
Na^+/K^+ type (D)	74
No dominant type (B)	16
Anions	
HCO_3^- type (E)	30
SO_4^{2-} type (F)	Nil
Cl^- type (G)	50
No dominant type (H)	20

interaction or rock domination are the main mechanisms regulating the global water chemistry. The Gibbs (1970) chart has been widely used to identify and understand hydrogeochemical processes that are important for determining the basis for improvements in the quality of groundwater. For anions ($\text{Cl}^-/(\text{Cl}^- + \text{HCO}_3^-)$) and cations ($\text{Na}^+ + \text{K}^+ /(\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$) of the groundwater samples, Gibbs (1970) suggested separate plotting of the groundwater samples against TDS. TDS versus $\text{Cl}^-/(\text{Cl}^- + \text{HCO}_3^-)$ and TDS versus $(\text{Na}^+ + \text{K}^+) /(\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$ were categorized into three zones: precipitation, rock dominance zone, and evaporation dominance zones. Figure 5 was plotted relating to $\text{Na}^+ + \text{K}^+$; $\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+}$ and Cl^- ; $\text{Cl}^- + \text{HCO}_3^-$ vs TDS to evaluate the mechanism that controls the groundwater chemistry with respect to the domains of atmospheric precipitation, rock weathering, evaporation and fractional crystallization. The results of the present study depicts the groundwater chemistry is significantly influenced by rock-water interaction.

Weathering of granite gneisses is a significant factor contributing to groundwater enriched with fluoride (Fig. 5) (Adimalla and Venkatayogi, 2017). Gibbs plot reveals that most of the groundwater belongs to rock dominance and later shifted to evaporation dominance influenced by dry climate. Figure: 5 depicts the chemistry of groundwater samples, which shows that the majority of samples lean toward rock dominance to evaporation dominance, suggesting an increase in Na^+ , Ca^{2+} , HCO_3^- , and Cations and a resulting higher TDS due to

water contamination brought on by influences from unsanitary conditions, agricultural fertilizers, and irrigation-return flows (SubbaRao, 2008). The distribution trend of the groundwater samples of this region suggests clearly that the groundwater quality was at first constrained by rock-water interaction and afterwards altered by anthropogenic exercises. Cation exchange reactions are the significant geochemical responses which influence the dissemination and occurrence of ions in groundwater can be recognized utilizing a connection between Na^+ and Cl^- ions. Ion exchange, anthropogenic and agricultural contamination presumably causes increased sodium concentration in a gneissic terrain (Guo and Wang, 2004). Increased Na^+ fixation as for Cl^- or decreased Na^+ as for Cl^- is the consequence of cation exchange responses (Salama, 1993; Rajmohan and Elango, 2004). In the typical ion exchange reaction, Ca^{2+} is held in the aquifer material and Na^+ is discharged to water. The overabundance of Na^+ created by ion exchange reaction is not adjusted by Cl^- however by alkalinity or SO_4^{2-} . Additionally, in the reverse ion exchange, Na^+ is held by aquifer materials and Ca^{2+} is discharged to water. For this situation, abundance Cl^- over Na^+ is adjusted by Ca^{2+} and Mg^{2+} . Subsequently, overabundance Na^+ over Cl^- or abundance Cl^- over Na^+ is a good sign for ion exchange reactions. In this case; excess Cl^- over Na^+ is balanced by Ca^{2+} and Mg^{2+} . Hence, higher Na^+ over Cl^- or higher Cl^- over Na^+ is a significant indication for ion exchange mechanisms observed in this region.

Weathering and ion exchange processes in the study area

The cation exchange is a significant natural phenomenon observed in the groundwater system which changes the groundwater hydrochemistry (Wu et al., 2014). Weathering and ion exchange mechanisms are the predominant solute inferring processes impact the convergence of groundwater chemistry. The ion exchange reactions of water and its aquifer material during residence or in movement process are assessed by Chloro-Alkaline Indices proposed by (Schoeller, 1967) as Schoeller index. The Chloro-Alkaline indices are computed using milliequivalent of Na^+ , K^+ , Cl^- , HCO_3^- , CO_3^{2-} , SO_4^{2-} and NO_3^- .

$$I = \text{CAI} = [\text{Cl}^- (\text{Na}^+ + \text{K}^+)] / \text{Cl}(1)$$

$$II = [\text{Cl}^- (\text{Na}^+ + \text{K}^+)] / \text{SO}_4^{2-} + \text{HCO}_3^- + \text{CO}_3^{2-} + \text{NO}_3^- (2)$$

Positive or negative value of CAI depends on replacement of Na^+ and K^+ of water with Mg^{2+} and Ca^{2+} in aquifer/soil or vice versa. If Na^+ & K^+ of water are replaced with Mg^{2+} & Ca^{2+} , in the rocks, positive values, depicting a base exchange reaction and is a direct reaction. The negative values of the CAI ratio denote chloro-alkaline disequilibrium and it is a cation - anion exchange mechanism. In the present study, eighty percent of groundwater samples with positive values inferring a base-exchange process while remaining twenty percent of total groundwater samples depict negative Schoeller index inferring cation-anion exchange mechanism.

Inter-elemental relations

Weathering of parent rocks is one of the major mechanisms controlling the concentrations of chemical constituents in the groundwater (Marghade et al., 2012). The relative proportion of various dissolved ions in groundwater depends, to a great degree, on their abundance in the host rocks/aquifer and their solubilities (Davraz and Özdemir, 2014). The plots delineated are helpful in understanding the weathering processes of rocks that control groundwater chemistry.

Ion exchange and cation adsorption are significant factors in regulating the ion ratios in groundwater, in addition to weathering and mineral dissolution. The $\text{Ca}^{2+} + \text{Mg}^{2+}$ vs. $\text{HCO}_3^- + \text{SO}_4^{2-}$ scatter

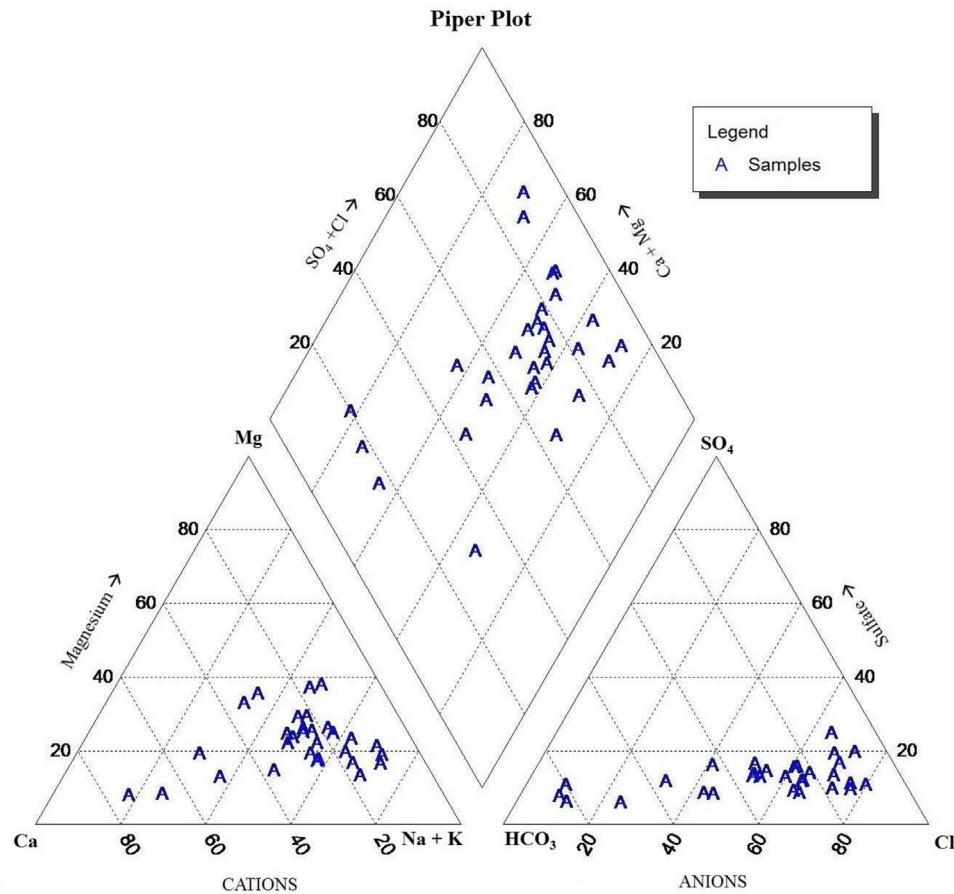


Fig. 4. Piper tri-linear diagram representing the chemical analysis of the study area.

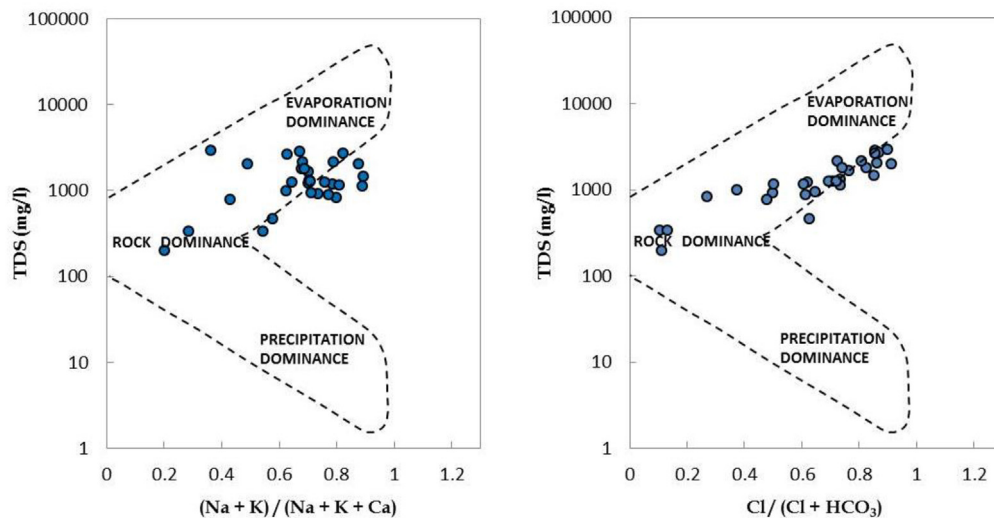


Fig. 5. Mechanisms controlling groundwater chemistry of the study area (after Gibbs, 1970).

map indicates that ion exchange is common throughout the research region.

The sources of dissolved constituents in groundwater reevaluated from the relative abundance of individual ions and inter-elemental correlation is as given below:

$\text{Ca}^{2+} + \text{Mg}^{2+}$ vs. $\text{HCO}_3^- + \text{CO}_3^{2-}$ plot (Fig. 6a) reveals that ninety percent of data points fall under the 1:1 equiline is due to depletion

of Ca^{2+} and Mg^{2+} of ionic exchange. In the $\text{Ca}^{2+} + \text{Mg}^{2+}$ vs total cations plot (Fig. 6b) depicts that hundred percent of data points fall below the 1:1 equiline indicates excess of total cations due to weathering of plagioclase, amphiboles, mica and K-feldspar etc. Similarly, eighty-four percent and sixteen percent of the total groundwater samples are noted above and fall underneath the equiline of Na^+ vs $\text{HCO}_3^- + \text{CO}_3^{2-}$ plot (Fig. 6c), which infers that the $\text{HCO}_3^- + \text{CO}_3^{2-}$ is

being contributed from the multiple sources, including the anthropogenic inputs. The plotting of chemical data of $\text{Na}^+ + \text{K}^+$ vs total cations (Fig. 6d) shows that fifty-five percent of the data points fall below and forty-five percent of the data points fall above the 1:1 equiline, signifies that the supply of total cations through rock weathering and/or soil salts stored by evaporation due to contribution of $\text{Na}^+ + \text{K}^+$ to the total cations (Singh, 2000; Rajmohan and Elango, 2004). The plot of $\text{Na}^+ + \text{K}^+$ vs $\text{Cl}^- + \text{SO}_4^{2-}$ (Fig. 6e) denotes eighty percent of groundwater samples fall below the equiline indicates that increase of $\text{Na}^+ + \text{K}^+$ (Fig. 6e), reflects the derivation of these ions from soil salts with pres-

ence Na_2SO_4 and K_2SO_4 (Sarin et al., 1989; Datta and Tyagi, 1996; SubbaRao, 2008). Plotting of Na^+ vs Cl^- (Fig. 6f) shows eighty percent of the samples fall below the equiline which shows that the release of Na^+ from the silicate weathering, which is an evidence of ion exchange reaction (Stallard and Edmond, 1983; Meybeck, 1987). Thus, the groundwater shows a dominance of HCO_3^- , CO_3^{2-} ions, as reported by Rogers (1989). The plotting of chemical data in Na^+ vs Ca^{2+} (Fig. 6g) shows that ninety percent of the samples fall below the equiline due to the enrichment of Na^+ rather than Ca^{2+} by the ionic exchange (SubbaRao, 2008). Besides cation exchange is additionally rec-

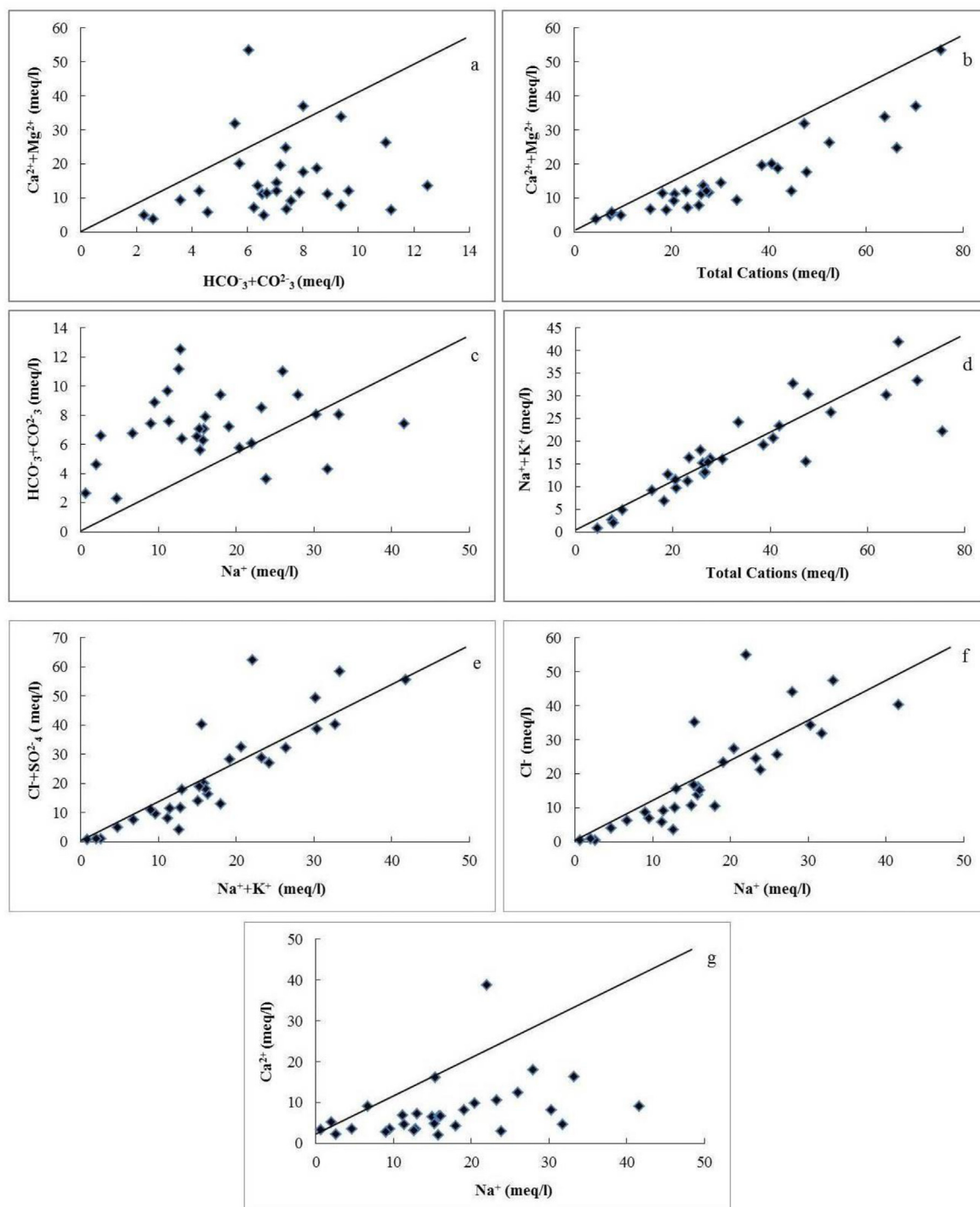


Fig. 6. a–g. Relation between major ions; a. $\text{Ca}^{2+} + \text{Mg}^{2+}$ and $\text{HCO}_3^- + \text{CO}_3^{2-}$ (meq/l). b. $\text{Ca}^{2+} + \text{Mg}^{2+}$ and Total Cations (meq/l). c. Na^+ and $\text{HCO}_3^- + \text{CO}_3^{2-}$ (meq/l). d. Total Cations and $\text{Na}^+ + \text{K}^+$ (meq/l). e. $\text{Na}^+ + \text{K}^+$ and $\text{Cl}^- + \text{SO}_4^{2-}$ (meq/l). f. Na^+ and Cl^- (meq/l). g. Na^+ and Ca^{2+} (meq/l).

Table 4
Correlation matrix of ions.

	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	Cl ⁻	SO ₄ ²⁻	CO ₃ ²⁻	HCO ₃ ⁻	NO ₃ ⁻	F ⁻
Ca ²⁺	1.00									
Mg ²⁺	-0.27	1.00								
Na ⁺	0.08	0.14	1.00							
K ⁺	-0.02	-0.29	0.52	1.00						
Cl ⁻	0.47	0.09	0.55	0.25	1.00					
SO ₄ ²⁻	0.07	0.25	0.03	-0.23	-0.07	1.00				
CO ₃ ²⁻	-0.61	0.11	-0.14	-0.01	-0.33	-0.22	1.00			
HCO ₃ ⁻	-0.50	-0.18	-0.02	0.23	-0.52	-0.44	0.39	1.00		
NO ₃ ⁻	-0.13	0.01	0.02	-0.21	-0.21	0.02	-0.14	-0.04	1.00	
F ⁻	-0.19	-0.16	-0.19	0.38	-0.06	-0.43	0.32	0.45	-0.24	1.00

Table 5
Factor Loadings of ions.

	F1	F2	F3	F4	F5
Ca ²⁺	-0.87	0.06	-0.21	0.02	-0.03
Mg ²⁺	0.39	0.23	-0.25	-0.22	-0.33
Na ⁺	-0.04	0.95	0.04	-0.15	0.25
K ⁺	0.01	0.36	0.07	0.32	0.87
Cl ⁻	-0.35	0.67	-0.51	0.39	-0.08
SO ₄ ²⁻	-0.03	-0.04	-0.32	-0.57	-0.01
CO ₃ ²⁻	0.66	-0.10	0.13	0.11	-0.02
HCO ₃ ⁻	0.36	-0.03	0.89	0.26	0.11
NO ₃ ⁻	0.05	-0.01	0.06	-0.41	-0.09
F ⁻	0.21	-0.16	0.20	0.64	0.25

Table 6
Percent contribution of each factor.

Factors	Eigen values	Percent contribution	Elements with high loadings
I	2.82	28	CO ₃ ²⁻
II	2.13	21	Na ⁺ , Cl ⁻
III	1.43	14	HCO ₃ ⁻
IV	1.13	11	F ⁻
V	0.82	8	K ⁺
Total		82	6

ognized by the decrease in the levels of Ca²⁺ concerning Na⁺ as denoted by Jankowski and Acworth (1993). Along these lines, it is apparent that cation exchange processes assume a considerable role as one of the geochemical forms governing the groundwater science of this area.

Factor analysis

Factor analysis is a notable multivariate technique, applied for most significant parts adding to the compound information structure and the interrelationships among factors (Lall and Sharma, 1996). It very well may be applied to control the contributions of normal substance enduring and different effects of chemical composition of groundwater. First, the correlation matrix, i.e., the array of correlation coefficients for all pairs of variables, was determined. After that, the matrix was diagnosed, and its principal components (eigenvectors) were determined. The so-called factor 1 is related to the higher Eigen value and can be applied to explain the greatest amount of variance in the data set. The second factor (orthogonal and uncorrelated with the first one) depicts most of the remaining variance, and so forth. Tables 4–6 summarize the results of the factor analysis of hydro chemical data.

In the entire study area four factors were identified which are controlling the groundwater chemistry. The contribution of the 1st factor is about 41.51% of the total variance and the Eigen value is 4.51, 2nd factor is about 13.29 % of the total variance and the Eigen value is

1.329, 3rd factor is about 9.43% of the total variance and the Eigen value is 0.943, 4th factor is about 9.01% of the total variance and the Eigen value is 0.90. The applicability of Factor Analysis depending upon the most significant first four factors reveals that about 73.85% total variance of the ions. The correlation coefficient denotes the degree of association that exists between two variables, one represented as a dependent variable. The systematic calculation of correlation coefficient among water quality variables impart indirect means for quick monitoring of water quality. The relationship between two variables is the correlation coefficient which shows how one variable predicts the other. Associated with correlation coefficient is *r*, which is the percentage of variance in the dependent variable, explained by the independent variable (Tiwari, 2011). The results of the correlation analysis are considered in the subsequent interpretation, high correlation coefficient (nearly 1 or -1) means a good relationship between two variables, and a correlation coefficient around zero means no relationship, positive values indicate a positive relationship while negative values of *r* indicate an inverse relationship. The correlation coefficients (*r*) of water quality parameters of groundwater of the study area were calculated and the values of correlation coefficients (*r*) are given in (Table 4).

The results of the statistical analysis given in Table 2 infers: (i) pH has positive correlation with CO₃²⁻, F⁻, K⁺; weak correlation with HCO₃⁻; negative correlation with EC, TDS, TH, Na⁺, Mg²⁺, Ca²⁺, Cl⁻, SO₄²⁻, HCO₃⁻, NO₃⁻ and F⁻. (ii) EC has positive correlation with TDS, TH, Na⁺, K, Mg²⁺, Ca²⁺, Cl⁻, SO₄²⁻; weak correlation with HCO₃⁻, NO₃⁻, K⁺ and F⁻; negative correlation with CO₃²⁻. (iii) TDS has positive correlation with TH, Na⁺, K⁺, Mg²⁺, Ca²⁺, Cl⁻, SO₄²⁻; weak correlation with HCO₃⁻, NO₃⁻, F⁻, CO₃²⁻; negative correlation with K⁺. (iv) TH has positive correlation with Na⁺, Mg²⁺, Ca²⁺, Cl⁻, SO₄²⁻; weak correlation with NO₃⁻, F⁻, HCO₃⁻; negative correlation with CO₃²⁻. (v) Na⁺ has positive correlation with Mg²⁺, Cl⁻, SO₄²⁻; weak correlation with NO₃⁻, F⁻, HCO₃⁻, K⁺, Ca²⁺; negative correlation with CO₃²⁻. (vi) K⁺ has weak correlation with F⁻, Mg²⁺, Cl⁻, SO₄²⁻, NO₃⁻, F⁻, CO₃²⁻, Ca²⁺; negative correlation with HCO₃⁻. (vii) Mg²⁺ has positive relation with Cl⁻, SO₄²⁻, HCO₃⁻; weak correlation with NO₃⁻, F⁻; negative correlation with Ca²⁺, CO₃²⁻ (viii) Ca²⁺ has positive rela-

tion with Cl^- , SO_4^{2-} ; weak correlation with HCO_3^- , NO_3^- ; negative correlation with F^- , CO_3^{2-} . (ix) Cl^- has positive correlation with SO_4^{2-} ; weak correlation with HCO_3^- , NO_3^- , F^- negative association with CO_3^{2-} . (x) SO_4^{2-} weak correlation with HCO_3^- , NO_3^- , F^- ; negative correlation with CO_3^{2-} . (xi) HCO_3^- weak correlation with NO_3^- , F^- . (xii) CO_3^{2-} weak relation with NO_3^- , F^- . (xiii) NO_3^- negative correlation with F^- .

Conclusions

Hydrochemical data reveals that the groundwater is slightly alkaline, very hard, high EC and TDS values are due to contamination of water, inappropriate drainage system, ion exchange and solubilisation process within the aquifers >500 mg/L, denoting non-potable category. In the case of anions, eighty percent of the samples exceeds the Cl^- safe limit of 250 mg/L, fifty-seven percent of the samples exceed the sulfate admissible limit of 200 mg/L, causing laxative effect in human, together with Na^+ or Mg^{2+} in drinking water, forty-five percent of samples are crossing the safe limit of NO_3^- . F^- content exceeds 1.5 mg/L in forty percent and below 0.6 mg/L in ten percent of total groundwater samples. As per Piper's trilinear diagram groundwater samples of the study area can be grouped into $\text{Na}^+ - \text{HCO}_3^- - \text{Cl}^-$ and $\text{Na}^+ - \text{Cl}^- - \text{HCO}_3^-$ water types. Anthropogenic activities cause the deviation of groundwater quality, further the rock-weathering and the dominance of evaporation zone indicates interaction between host rock of aquifer and groundwater, which is the remarkable process governing the groundwater chemistry and groundwater quality, initially controlled by rock-water interaction, subsequently modified by anthropogenic activities supported by Gibbs diagram. Based on Schoeller Index, twenty percent of total groundwater samples show negative values because of cation-anion exchange and eighty percent show positive values inferring a base exchange reactions. Inter elemental relations, the rock-weathering is the significant mechanism controlling the groundwater chemistry. Based on the factor analysis results, four factors are contributing to 74 percent of groundwater contamination. This work is of enormous societal advantage with regard to existing human health hazards in this region and has immediate applicability to such industrially inhabited areas somewhere else.

Data availability Every related data and material is given in the main paper.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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